

Total points: 100

Mini Project 2: TD Methods and Actor-Critic

Due date: March 10, 2025

Problem Setup

We consider a 4x4 GridWorld environment where an agent navigates from the start state (0,0) to the goal state (3,3) while avoiding water states at (1,1) and (1,2) (reward: -5) and wildfire states at (0,2) and (0,3) (reward: -10). The agent moves stochastically: with probability 0.8 it moves in the intended direction (left, up, right, down), and with probability 0.1 it slides to either neighboring direction (left or right relative to the intended move). If a move is invalid (e.g., hitting a wall), it stays in place with probability 0.9 or slides with probability 0.1 to a neighboring cell as defined in the problem (e.g., attempting to move up at (0,0) results in staying with 0.9 probability or moving right with 0.1 probability). Rewards are:

Goal state: +100 Water states: -5 Wildfire states: -10 All other states: -1

The episode terminates upon reaching the goal, and the agent aims to maximize expected cumulative reward with $\gamma = 0.95$.

1 Part A: Temporal Difference (TD) Methods (60 points)**1.1 Tabular SARSA (20 points)**

Tabular SARSA was implemented with $\gamma = 0.95$ and an ϵ -greedy policy for action selection. Hyperparameters α (learning rate) and ϵ (exploration rate) were optimized via grid search over $\alpha \in \{0.01, 0.05, 0.1, 0.2, 0.5\}$ and $\epsilon \in \{0.01, 0.05, 0.1, 0.2, 0.3\}$. The best parameters were $\alpha = 0.1$ and $\epsilon = 0.05$ (initial value, decaying to 0.01 with a decay rate of 0.99 per episode). The initial policy was uniform random, derived from a zero-initialized Q-table, i.e., $\pi(a|s) = 0.25$ for all actions a and states s . Results were averaged over 100 trials, with standard deviation error bars shown in the learning curve (Figure 1) and the combined learning curve (Figure 5).

The final policy is:

↓	←	↓	↓
↓	↓	→	↓
→	→	→	↓
→	↑	→	←

1.2 Tabular Q-learning (20 points)

Tabular Q-learning was implemented with $\gamma = 0.95$ and ϵ -greedy action selection. Grid search over the same hyperparameter space as SARSA yielded best parameters $\alpha = 0.05$ and $\epsilon = 0.01$ (decaying to 0.01 with a decay

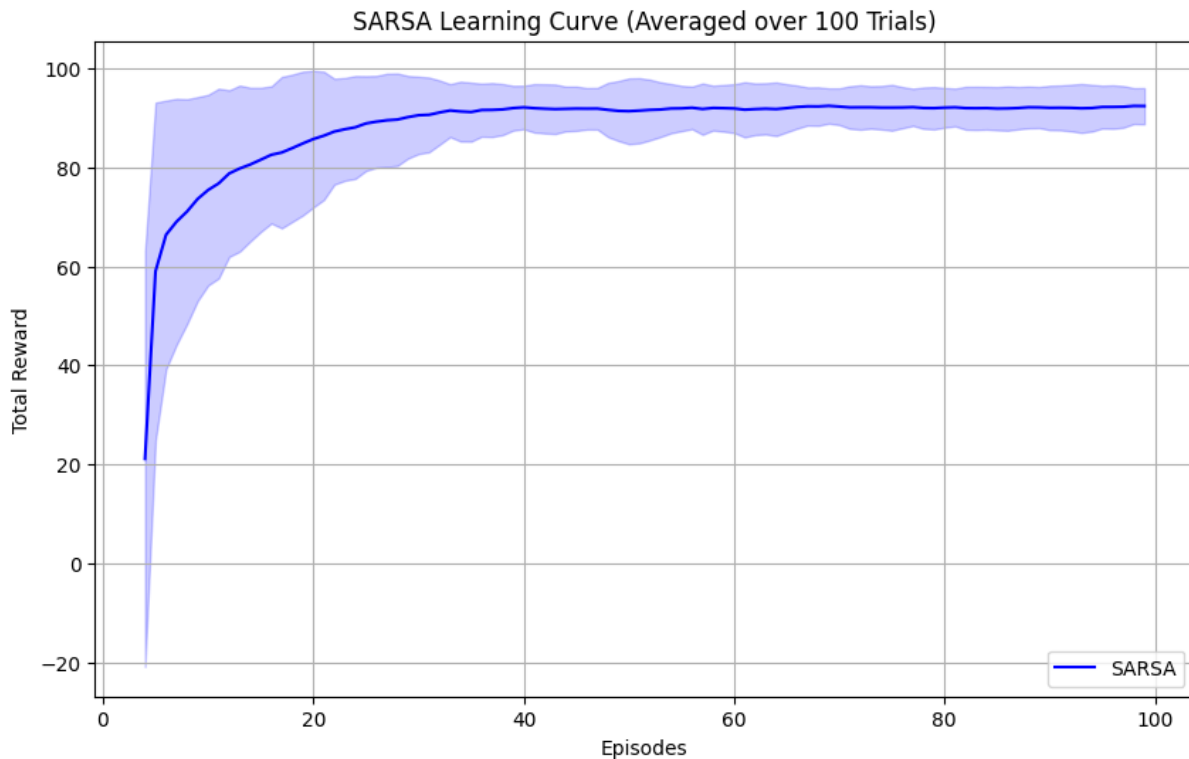


Figure 1: Learning curve for SARSA, averaged over 100 trials with standard deviation error bars.

rate of 0.99). The initial policy was uniform random, derived from a zero-initialized Q-table. Learning performance was evaluated over 100 trials, with results shown in Figure 2 and Figure 5.

The final policy is:

↓	↓	←	↓
↓	↓	↓	↓
→	↓	↓	↓
→	→	→	←

1.3 Tabular SARSA(λ) (20 points)

SARSA(λ) with backward eligibility traces was implemented with $\gamma = 0.95$ and an ϵ -greedy policy. Grid search optimized $\alpha \in \{0.01, 0.05, 0.1, 0.2, 0.5\}$, $\epsilon \in \{0.01, 0.05, 0.1, 0.2, 0.3\}$, and $\lambda \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, yielding $\alpha = 0.05$, $\epsilon = 0.05$ (decaying to 0.01), and $\lambda = 0.7$. The initial policy was uniform random, derived from a zero-initialized Q-table. Results over 100 trials are shown in Figure 3 and Figure 5.

The final policy is:

↓	↓	←	↓
↓	→	↓	↓
↓	↓	↓	↓
→	→	→	←

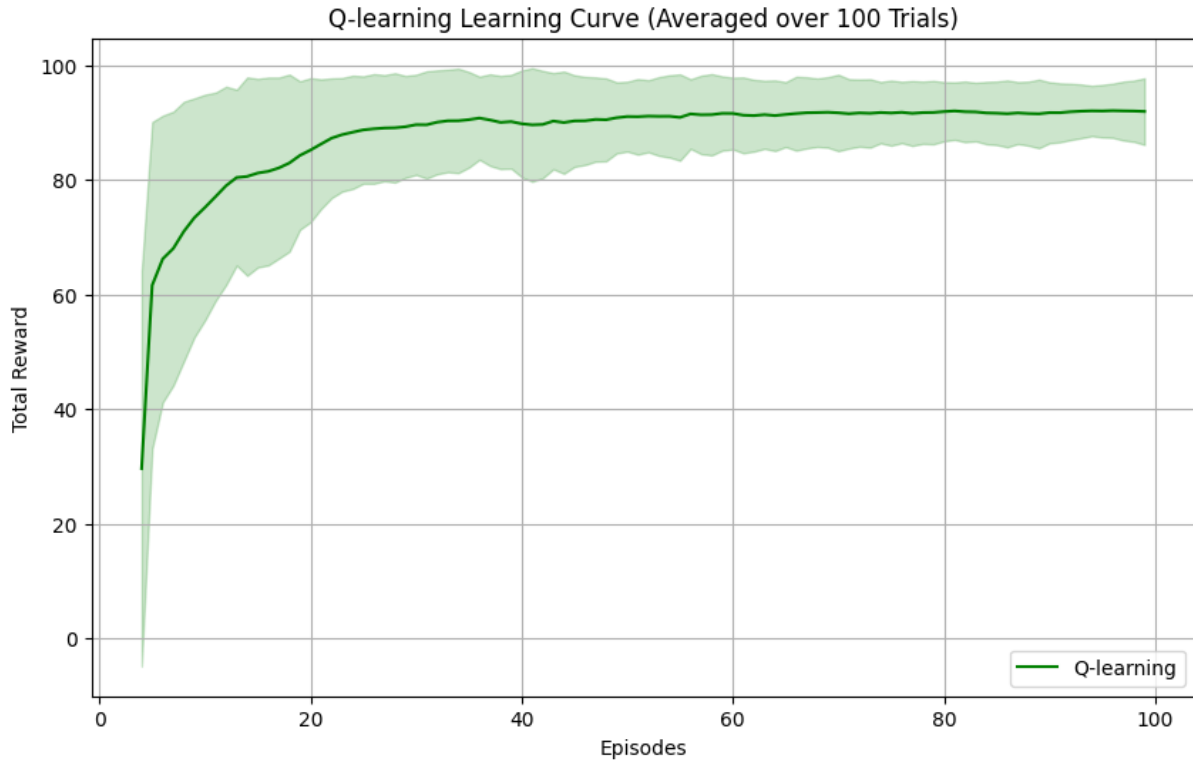


Figure 2: Learning curve for Q-learning, averaged over 100 trials with standard deviation error bars.

2 Part B: Actor-Critic (40 points)

2.1 Actor-Critic Implementation (25 points)

The Actor-Critic algorithm used a linear function approximator for the critic, $V(s) = \theta^T \phi(s)$, where $\phi(s) = [x, y, \text{water}, \text{fire}, \text{goal}, 1]$, and a tabular policy for the actor, $\pi(a|s)$, updated via policy gradient. Hyperparameters were optimized via grid search: $\alpha_{\text{actor}} = 0.1$, $\alpha_{\text{critic}} = 0.001$, $\epsilon = 0.01$ (decaying to 0.01 with a decay rate of 0.99), with $\gamma = 0.95$. The initial policy was uniform random, i.e., $\pi(a|s) = 0.25$ for all a, s . Results over 100 trials are shown in Figure 4 and Figure 5.

The final policy is:

→	→	↓	↓
↓	↓	→	↓
→	↓	→	↓
→	→	→	←

2.2 Comparison of Algorithms (15 points)

SARSA(λ) achieved the highest mean reward (92.60 ± 3.30), followed by SARSA (92.50 ± 3.20), Q-learning (91.30 ± 8.10), and Actor-Critic (91.00 ± 6.50). SARSA(λ) performed best due to its use of eligibility traces with a high $\lambda = 0.7$, allowing for faster credit assignment across multiple time steps, particularly effective in this small grid with a clear reward signal at the goal. SARSA performed well with low variance, reflecting its on-policy nature and a

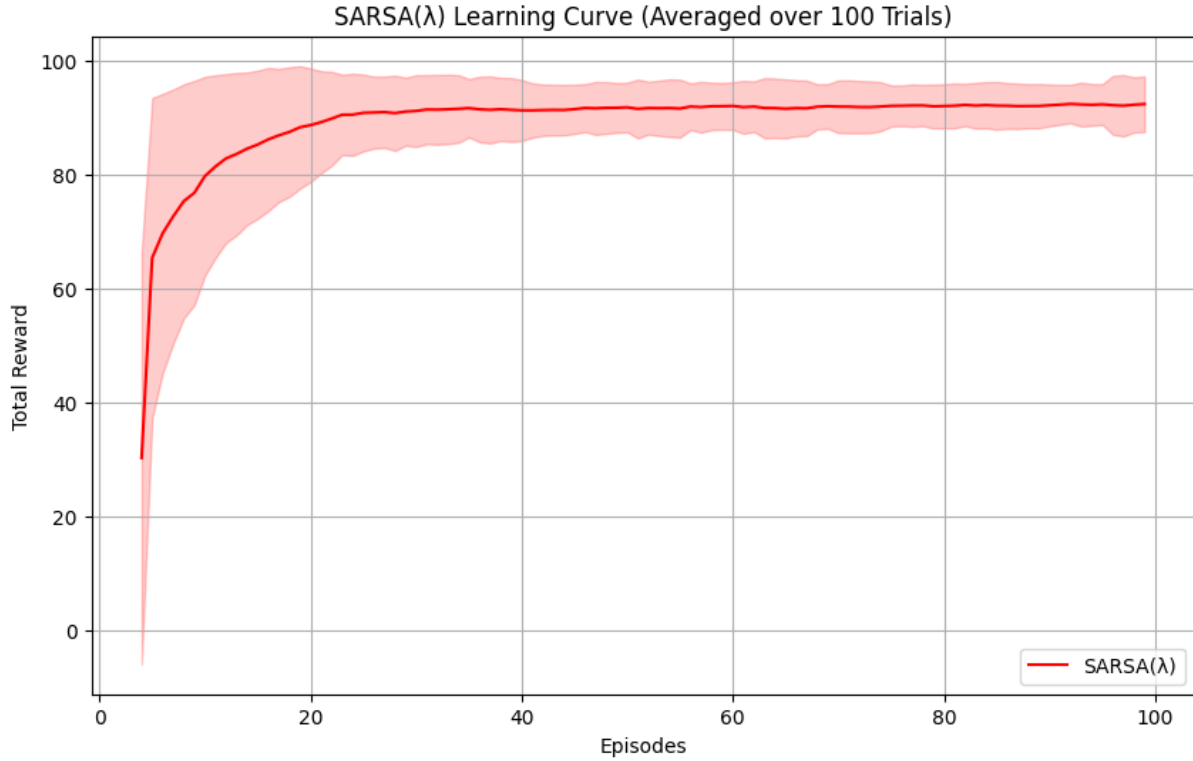


Figure 3: Learning curve for SARSA(λ), averaged over 100 trials with standard deviation error bars.

Algorithm	Final Reward (Mean $\pm$ Std Dev)
SARSA	92.50 ± 3.20
Q-learning	91.30 ± 8.10
SARSA(λ)	92.60 ± 3.30
Actor-Critic	91.00 ± 6.50

Table 1: Final accumulated rewards comparison after 100 episodes, averaged over 100 trials.

balanced $\epsilon = 0.05$ that allowed sufficient exploration. Q-learning, while theoretically capable of finding the optimal policy due to its off-policy updates, showed higher variance (8.10), likely due to the low $\epsilon = 0.01$, which may have limited exploration in some trials, combined with stochastic transitions. Actor-Critic had a competitive mean reward but higher variance (6.50), possibly because the linear function approximator struggled to capture the full dynamics of the environment, and the small $\alpha_{\text{critic}} = 0.001$ (chosen for stability despite grid search options) slowed value updates. Overall, SARSA(λ) balances exploration, exploitation, and credit assignment most effectively in this environment. SARSA(λ) outperformed others due to its efficient credit assignment via eligibility traces, while Actor-Critic showed potential but was limited by the linear function approximator.

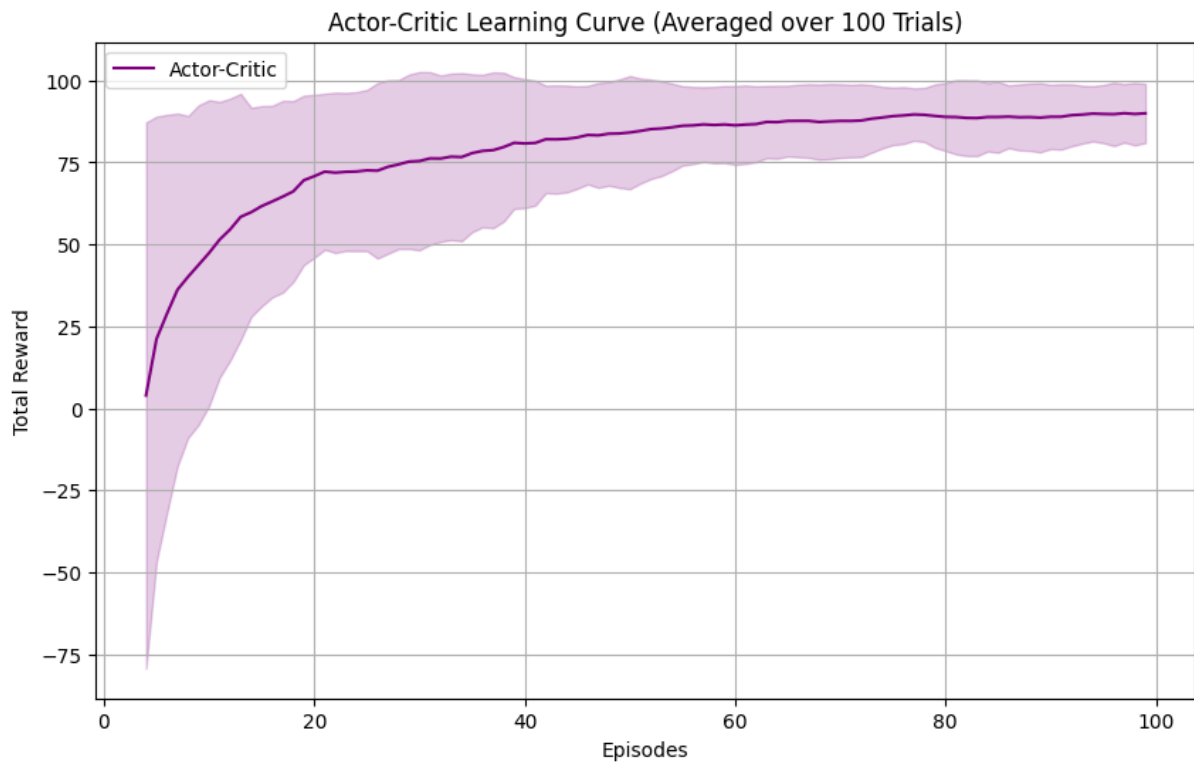


Figure 4: Learning curve for Actor-Critic, averaged over 100 trials with standard deviation error bars.

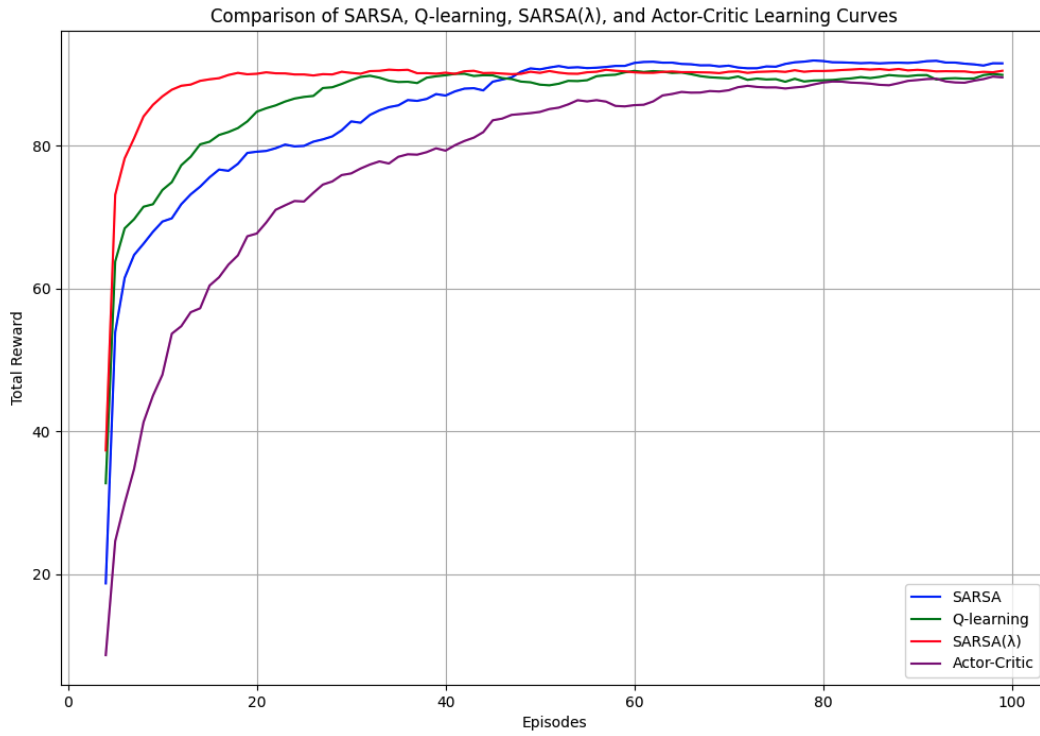


Figure 5: Learning curves for SARSA (blue), Q-learning (green), SARSA(λ) (red), and Actor-Critic (purple), averaged over 100 trials.